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Personalization - A Taxonomy

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ABSTRACT

Personalization is here defined as a process of changing a system to increase its personal relevance. This may have a work or social motivation. A taxonomy of motivations is developed and illustrated by application to mobile phones and e-commerce Web pages.

Keywords

Personalization, customization, e-commerce, self-presentation, agents, social interfaces.

INTRODUCTION

There is much current interest in the personalization of products be they software or hardware. Thus mobile phones are now sold with replaceable coloured covers, e-commerce sites learn a users preferences and word processors allow you to customize the menus and tool bars.

Personalization is defined here as a process that changes the functionality, interface, information content, or distinctiveness of a system to increase its personal relevance to an individual. The effect of the changes should persist across sessions [1]. If this were not so, almost anything we do when using a computer, e.g., entering data, could be regarded as personalization.

Personalization can be system or user-initiated, the former often being described as customization. It is important to view this as a dimension rather than as a dichotomy as often both the system and the user participate in the process. For instance, the system may recognise that some aspects of the user's interaction with the software could be optimised but the change does not take place until the user has authorised it.

The next section sets out a taxonomy of personalization that focuses on the user's work and social motivations for personalization. The alternative would have been to focus on the nature of the changes made to the system. The advantage of our user-centred taxonomy is that it is applicable to a wide range of products. To illustrate this the third section applies it to examples of personalization to be found in mobile phones and e-commerce web sites.

WHY PERSONALIZE?

Motivations for personalization can be divided into those that are primarily to facilitate the work and those that are primarily to accommodate social requirements. The former motivational category contains three subcategories:

enabling access to information content, accommodating work goals, and accommodating individual differences; the latter contains two subcategories: *eliciting an emotional response* and *expressing identity*.

Work related motivations to personalize

First, the user may want to personalize to *enable access to information content*. This would be the case when creating a bookmark or when indicating the type of information that should be displayed on the starting page of a portal. This aspect of motivation is particularly important in the World Wide Web where the amount of information exceeds our processing capacities.

The second work-related reason to personalize would be to *accommodate work goals*. For example, Mackay found that avoiding repetition or stopping something that is annoying or slow were triggers for customizing software among employees at MIT [1].

Third, personalization may be to *accommodate individual differences*. An instance of this would be providing access for people with visual disabilities. More exotically, Reeves and Nass [2] suggest that personality influences users' preferences towards computers. Dominant individuals seem to prefer a computer with a dominant dialogue style whereas submissive users prefer a submissive computer.

Socially related motivations to personalize

Scheiberg [3] has suggested that people personalize their physical workspace in order to *elicit emotional responses*. Similar effects might be expected when personalizing a product. Perhaps, a user chooses a particular ringing tone for a mobile phone because it is associated with emotions such as pleasure or excitement.

Personalized systems may be useful in defining, maintaining and *expressing the identity* of the user. Goffman [4] uses the metaphor of a stage performance to explain self-presentation. Thus a product may be thought of as a "prop" or "expressive equipment" employed by the individual during his or her performance. An important part of social identity is the group or groups one associates with. Personalization can be motivated by the need to do this. For example, a peer group might be attaching stickers to their phones in order to identify themselves with that particular group.

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APPLYING THE MODEL TO EXISTING SYSTEMS

What follows, is an attempt to show how the current model applies to 'real world' examples of human computer interaction. The model of personalization will be imposed upon mobile phones and e-commerce Web pages. These were chosen as the products differ greatly from each other and are both familiar examples of domains where personalization is currently utilised. In each example, the contributions of both the user and the system to the process are described and the degree of initiation is determined (5-point scale where 1 refers to extreme user-initiation and 5 to extreme system-initiation). The motivational forces and objects of change are also included in the examples.

Examples of personalization - Mobile phone

Function 1: Appearance

Process: The user chooses a colour cover or attaches a sticker on the phone; the change persists in the system.

Degree of initiative (user versus system): 1.

Possible motivations: Identity (e.g., promoting a 'wild' self-image), emotional response (e.g., attachment to the phone).

What is changed: Distinctiveness.

Function 2: Personal ringing tone

Process: The user selects, orders, or composes the tone; the system memorises the change.

Degree of initiative (user versus system): 1.

Possible motivations: Individual differences (e.g., choose a tone that can be easily heard), identity, emotional response

What is changed: Interface and distinctiveness.

Function 3: Caller recognition

Process: The user couples an identifier (ringing tone, image, or name) with the event of a specific person or a person belonging to a particular group calling the user; the coupling is memorised by the system.

Degree of initiative (user versus system): 1.

Possible motivations: Access to information, individual differences, emotional response, and identity.

What is changed: Interface, information content, and distinctiveness.

Examples of personalization - E-commerce Web pages

Function 1: Making the interface suitable for visually impaired

Process: The user enlarges the font size or increases the contrast of the front; the system memorises the change.

Degree of initiative (user versus system): 1.

Possible motivations: Individual differences.

What is changed: Interface.

Function 2: Receiving information of the area of interest

Process: The user indicates which areas she is interested in; the system memorises this and will notify the user about items that fall within these defined categories (e.g., personalized news services, such as InfoComm.ca).

Degree of initiative (user versus system): 3.

Possible motivations: Access to information.

What is changed: Information content.

Function 3: Recommending products on the basis of preferences of similar-minded consumers

Process: The user submits personal information (e.g., feedback or ratings) to the system; the system uses a technology such as collaborative-based filtering [5] to recommend products to the user. Services such as Amazon.com and Moviecritic.com utilise this feature.

Degree of initiative (user versus system): 4.

Possible motivations: Access to information, identity (the recommendations may support the user's self-image), emotional response (the user is positively surprised about the relevance of the recommendations).

What is changed: Distinctiveness, information content.

Both of the above domains provide great opportunities for creating new personalization features. Techniques that are currently promising include, e.g., agent technologies [5] and affective computing [6]. An interesting instance of personalization takes place when both work and socially related factors trigger the change, such as in function 3 in the above examples. In these cases, the instrumental and symbolic values of the device are combined to make this process more complex and motivating.

DISCUSSION AND FUTURE WORK

The above represents an initial attempt at a framework for thinking about personalization. In this way we hope to increase consensus regarding the meaning of the term. Also the taxonomy may be used to identify possibilities for personalization when designing or enhancing a product.

Our plans for the future include field studies of how people use current facilities for personalization and experiments with new forms of personalization. This should enable a better understanding of how and why individuals personalize.

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